

Chapter 4 Study Sheet

Transformations — everything you need on one page.

Transformation	Rule	Rigid motion?
Translation along $\langle a, b \rangle$	$(x, y) \rightarrow (x + a, y + b)$	Yes
Reflection in the x-axis	$(a, b) \rightarrow (a, -b)$	Yes
Reflection in the y-axis	$(a, b) \rightarrow (-a, b)$	Yes
Reflection in $y = x$	$(a, b) \rightarrow (b, a)$	Yes
Reflection in $y = -x$	$(a, b) \rightarrow (-b, -a)$	Yes
Reflection in $x = k$	$(a, b) \rightarrow (2k - a, b)$	Yes
Reflection in $y = k$	$(a, b) \rightarrow (a, 2k - b)$	Yes
Rotation 90° about O	$(a, b) \rightarrow (-b, a)$	Yes
Rotation 180° about O	$(a, b) \rightarrow (-a, -b)$	Yes
Rotation 270° about O	$(a, b) \rightarrow (b, -a)$	Yes

The Three Rigid Motions

- Translation (4.1): slides every point the same distance in the same direction, along a vector $\langle a, b \rangle$.
- Reflection (4.2): flips a figure over a line. The line of reflection is the perpendicular bisector of every segment joining a point to its image.
- Rotation (4.3): turns a figure about a fixed center. All rotations are counterclockwise unless you are told otherwise.
- All three preserve length and angle measure, so in every case the image is CONGRUENT to the preimage.

Working Backward and Composing

- To find a PREIMAGE from an image, undo the rule: subtract what you added, and reverse the switch or the sign change.
- Two translations in a row: just add the vectors. $\langle a, b \rangle$ then $\langle c, d \rangle$ is the single rule $(x, y) \rightarrow (x + a + c, y + b + d)$.
- A glide reflection is a translation followed by a reflection in a line PARALLEL to the translation vector.
- Order usually matters. Do each step in the order it is written, and label the first image with ' and the second with ''.

Symmetry

- Line symmetry: a reflection in some line maps the figure onto itself. That line is a line of symmetry.
- Rotational symmetry: a rotation of 180° or less about the center maps the figure onto itself.
- A regular n-gon has n lines of symmetry and rotational symmetry every $360^\circ/n$ degrees.
- Careful: a rectangle that is not a square has 2 lines of symmetry, NOT 4 — its diagonals are not lines of symmetry.
- A parallelogram that is not a rectangle or rhombus has 0 lines of symmetry but DOES have 180° rotational symmetry.